

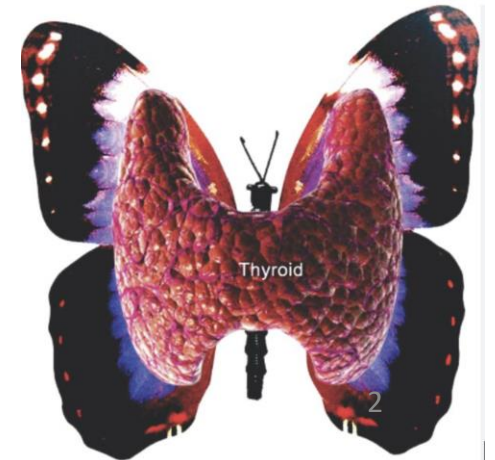


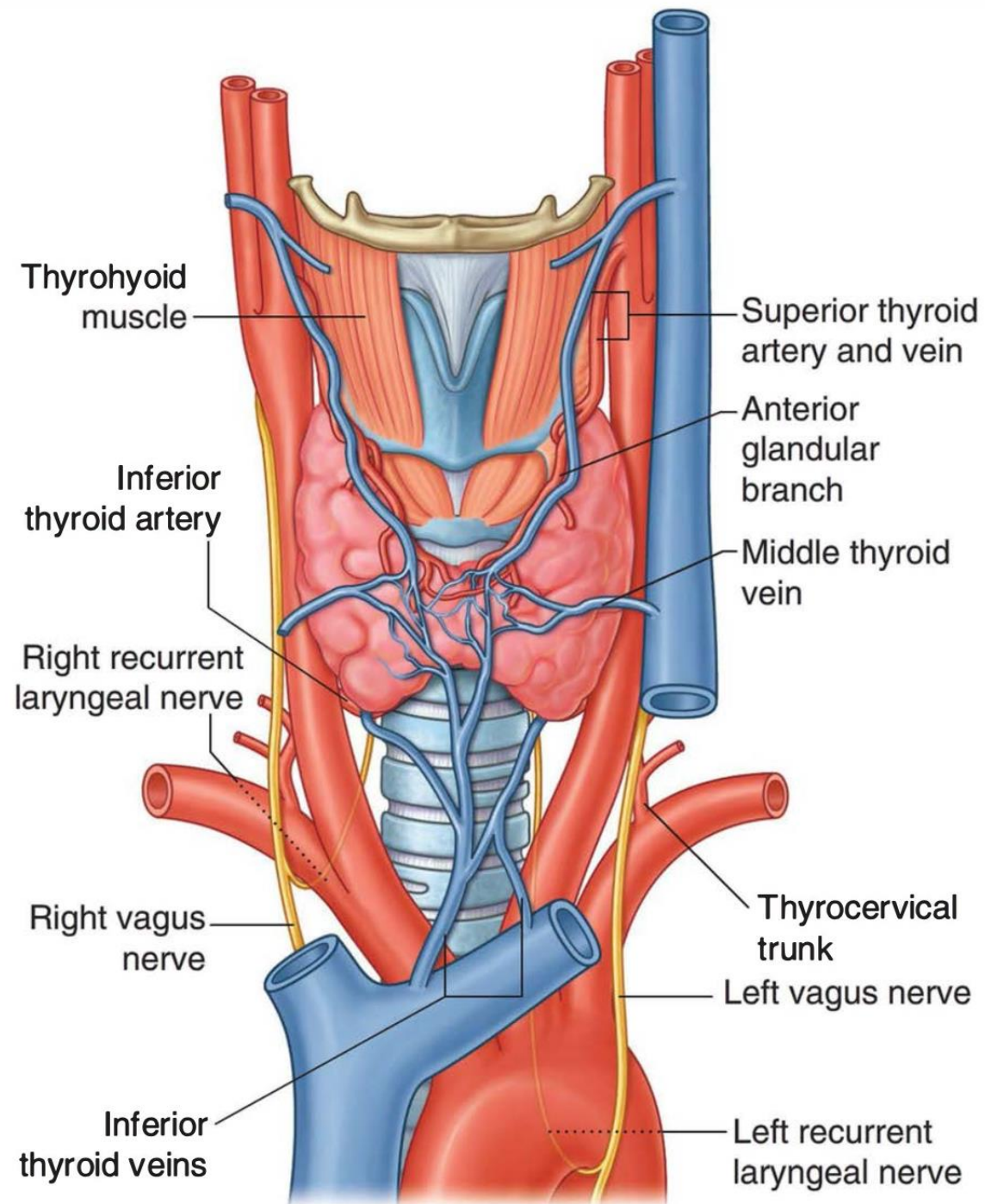
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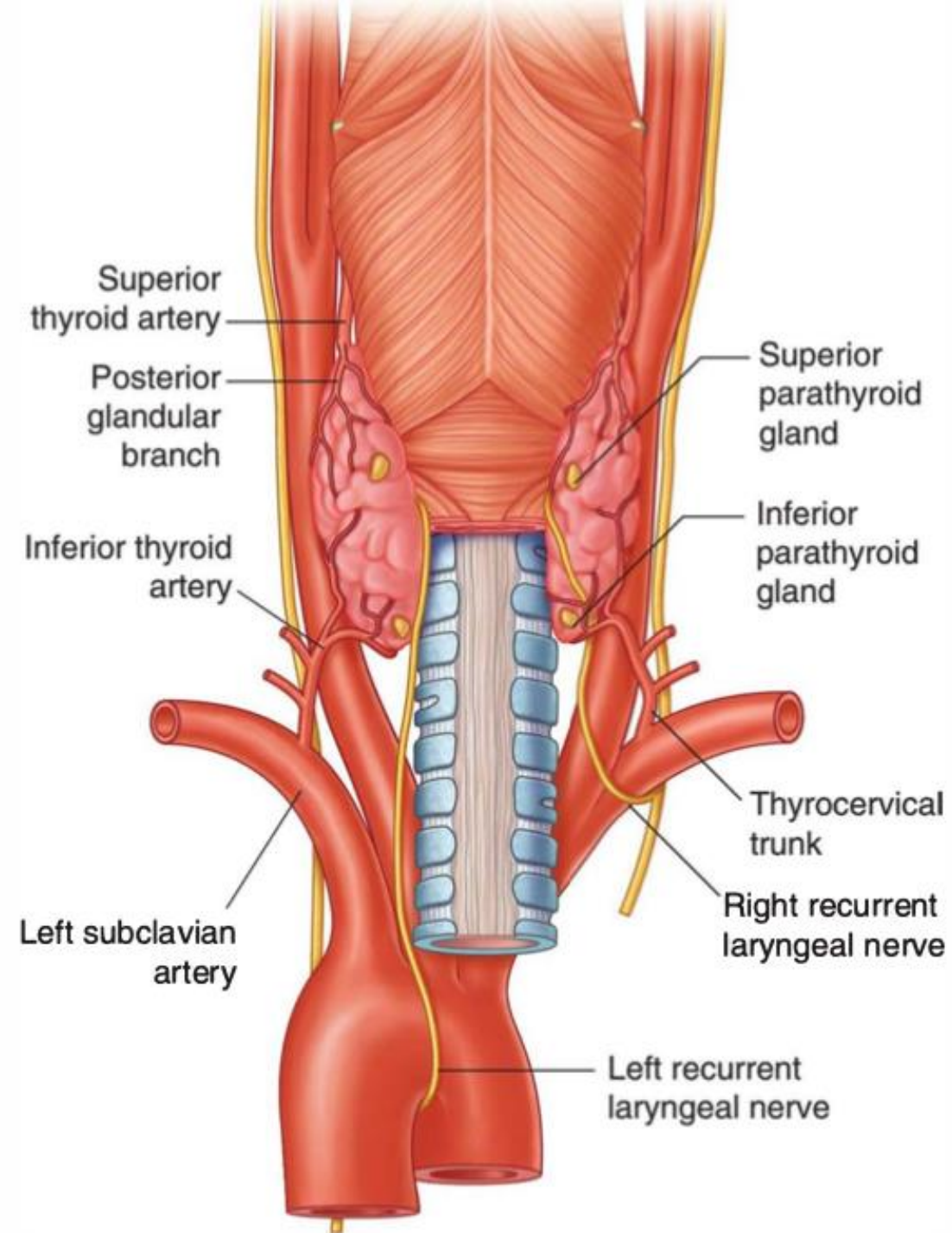
Thyroid diseases

Assistant professor
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- The thyroid gland is a butterfly-shaped endocrine gland located at the front of the neck below the thyroid cartilage.
- The normal weight of the thyroid gland in adults is between 10 and 20 grams.
- It is slightly larger and heavier in men than in women.



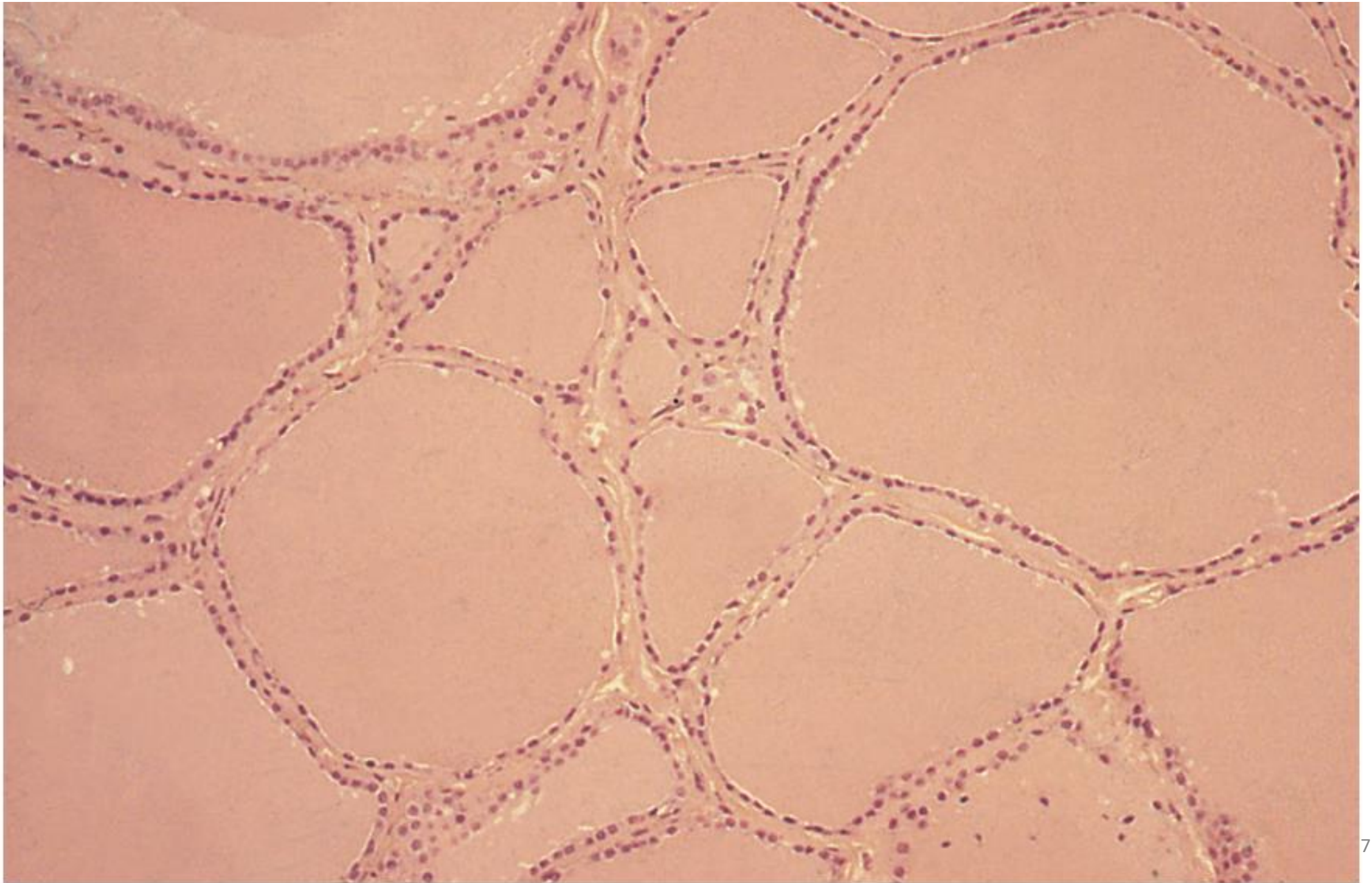






- The thyroid gland produces two main hormones
 1. Thyroxine (T4).
 2. Triiodothyronine (T3).

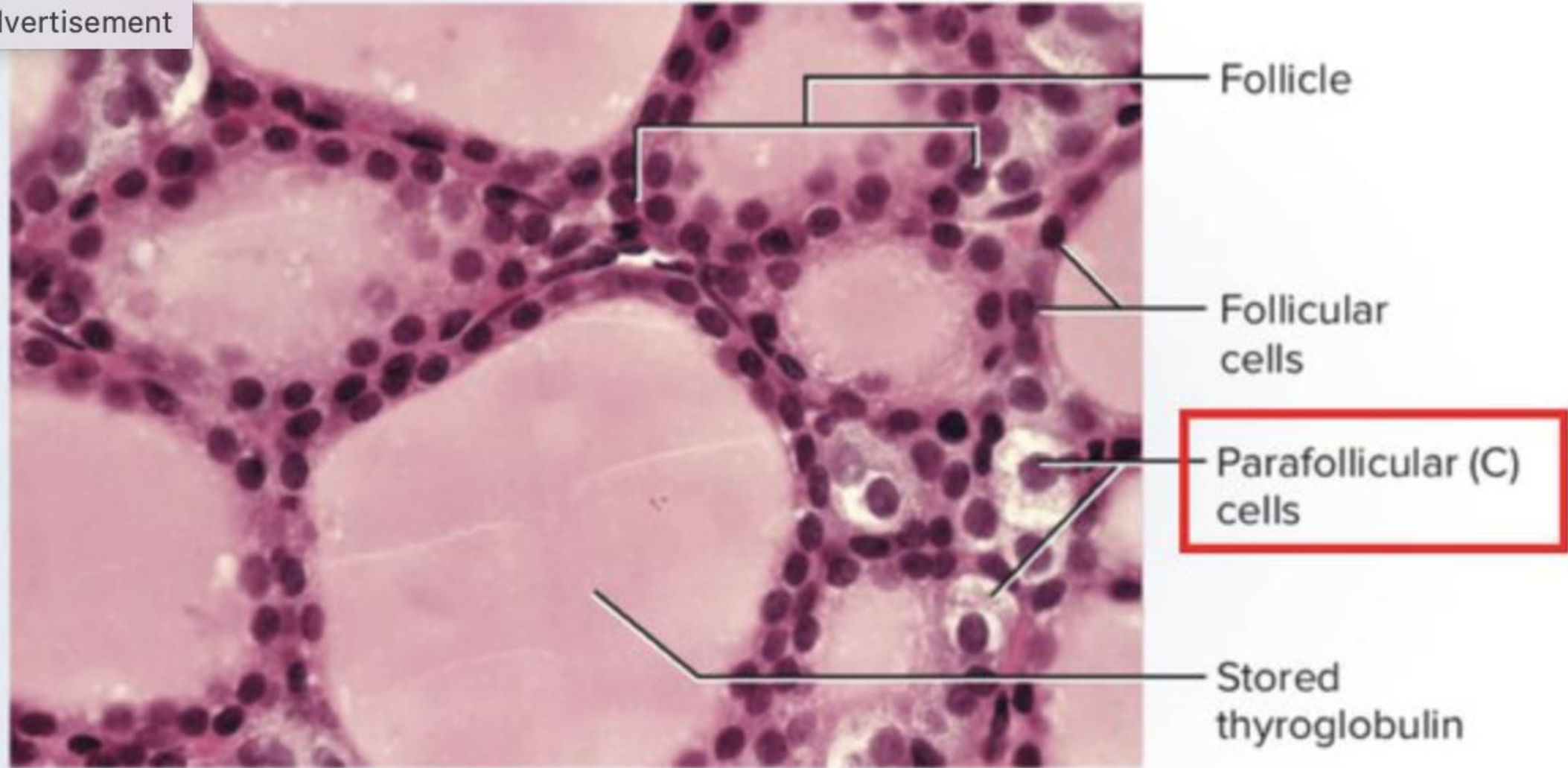
It's controlled by TSH from the anterior pituitary gland.





Calcitonin is secreted by parafollicular cells in the thyroid gland it's stimulated by hypercalcemia, and it lowers serum calcium levels.

Advertisement



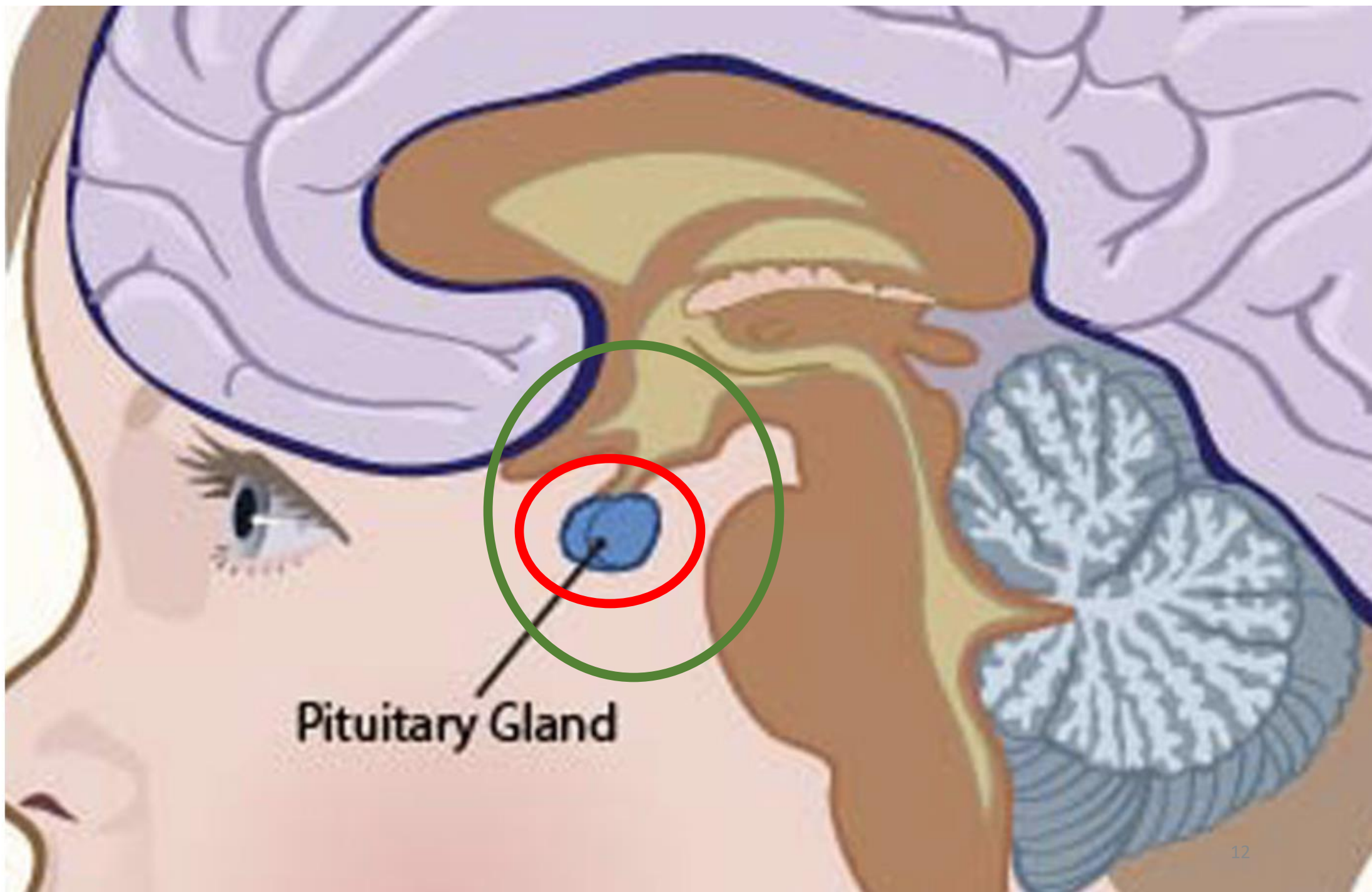


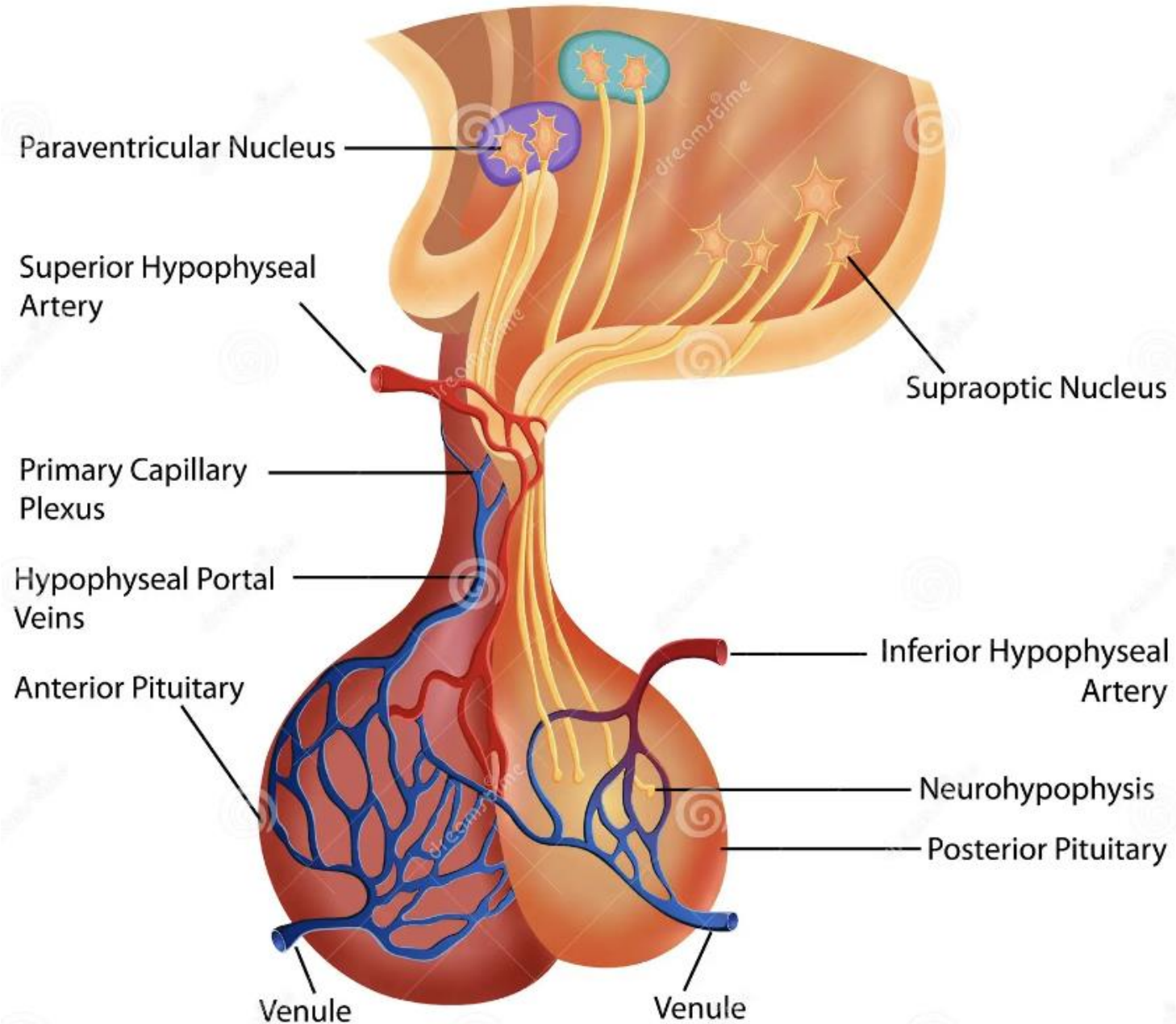
The pituitary–thyroid axis

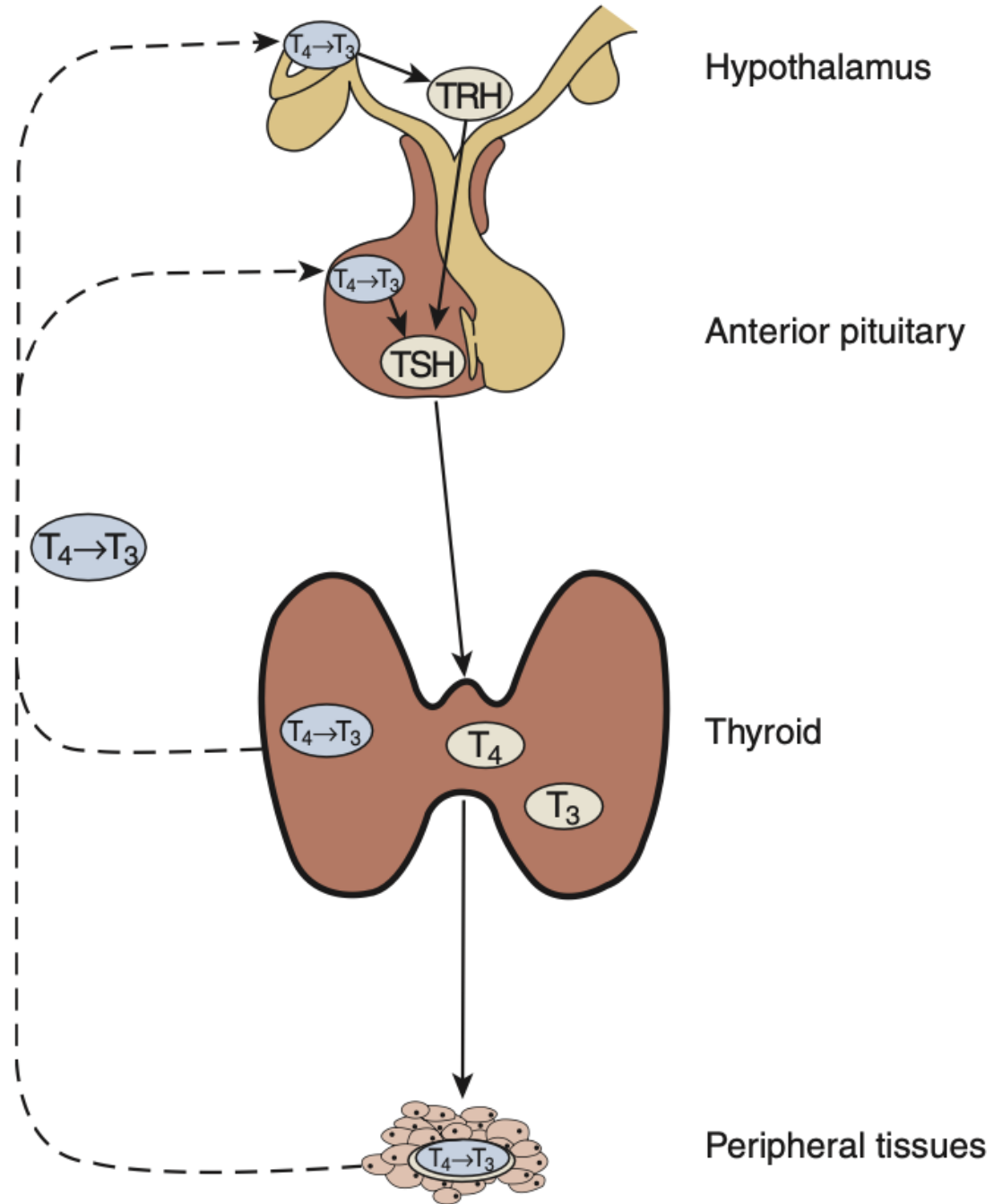
- Synthesis and release of thyroid hormones from the thyroid is controlled by thyroid-stimulating hormone (TSH) from the anterior pituitary.
- Secretion of TSH depends upon the level of circulating thyroid hormones and is modified in a negative feedback manner.



- In hyperthyroidism, TSH production is suppressed, whereas in hypothyroidism, it is stimulated.
- Regulation of TSH secretion also results from the action of thyrotrophin-releasing hormone (TRH) produced in the hypothalamus.









Hyperthyroidism

- It's an abnormally high thyroid hormone.
- Symptoms of hyperthyroidism can include weight loss, increased appetite, anxiety, irritability, heat intolerance, rapid heart rate, and tremors.
- Graves disease is associated with eye signs also.

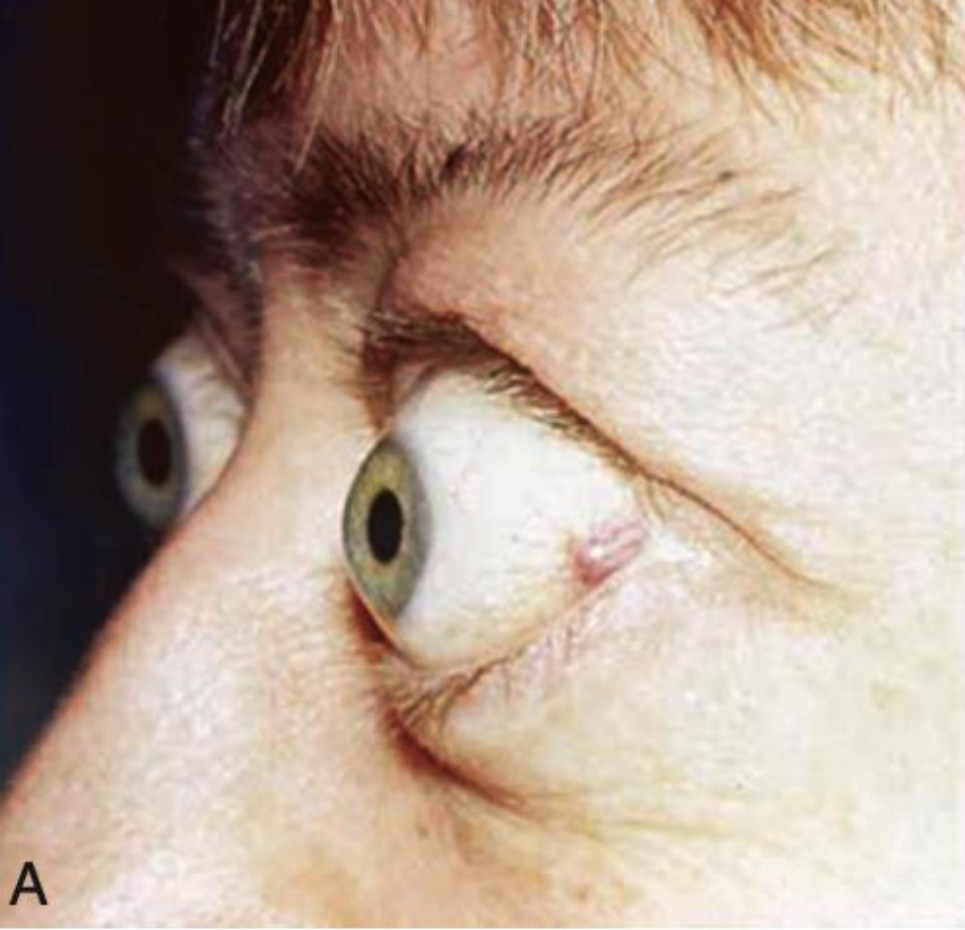


Causes of Hyperthyroidism :

Primary thyroid hyperfunction

1. Graves disease
2. Toxic multinodular goiter
3. Toxic adenoma

- Graves ophthalmopathy is characterized by edema, inflammation of the extraocular muscles, and an increase in orbital connective tissue and fat.
- Ophthalmopathy include
 1. Eyelid retraction.
 2. Proptosis.
 3. Periorbital edema.
 4. Chemosis.







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- Secondary thyroid hyperfunction
- Pituitary adenoma—TSH secretion



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- Without thyroid hyperfunction

Hormonal leakage—subacute thyroiditis

Iatrogenic (overdosage of thyroid hormone)



Hypothyroidism

- This condition occurs when the thyroid gland does not produce enough thyroid hormone.
- Symptoms include
 1. Fatigue.
 2. Decrease appetite.
 3. Weight gain.
 4. Constipation.
 5. Sensitivity to cold.
 6. Dry skin.





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Causes of Hypothyroidism (congenital hypothyroidism, myxedema)





Primary atrophic hypothyroidism

- Insufficient amount of thyroid tissue
- Destruction of tissue by autoimmune process: Hashimoto thyroiditis (atrophic and goitrous), Graves disease—end-stage
- Destruction of tissue by iatrogenic procedures: I^{131} therapy, surgical thyroidectomy, external radiation to the thyroid gland



- Defects of thyroid hormone biosynthesis

1. Congenital enzyme defects
2. Iodine deficiency or excess
3. Agenesis or dysplasia



Secondary hypothyroidism

- Pituitary
- Panhypopituitarism (neoplasm, irradiation, surgery)
- Isolated TSH deficiency

Hypothalamic causes

- Congenital



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- Transient hypothyroidism
- Silent and subacute thyroiditis
- Thyroxine withdrawal



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Thyroiditis

- It's inflammation of the thyroid gland.



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Thyroiditis

- Causes of Thyroiditis

1. Acute suppurative.
2. Subacute painful.
3. Subacute painless.
4. Hashimoto.
5. Chronic fibrosing (Riedel).



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Tumors

Type (Histologic)	Frequency (%)	10-Year Survival Rate (%)
Differentiated—papillary	75–80	>90
Differentiated—follicular	8–10	80
Differentiated—Hürthle cell	1	70
Anaplastic	1–5	<2
Medullary	5–8	40
Lymphoma	1–5	45
Metastases to the thyroid	<1	Determined by primary



Investigations

- Thyroid disorders can be diagnosed with a blood test to check thyroid hormone levels.
- Thyroid function test
 - T_3
 - T_4
 - TSH



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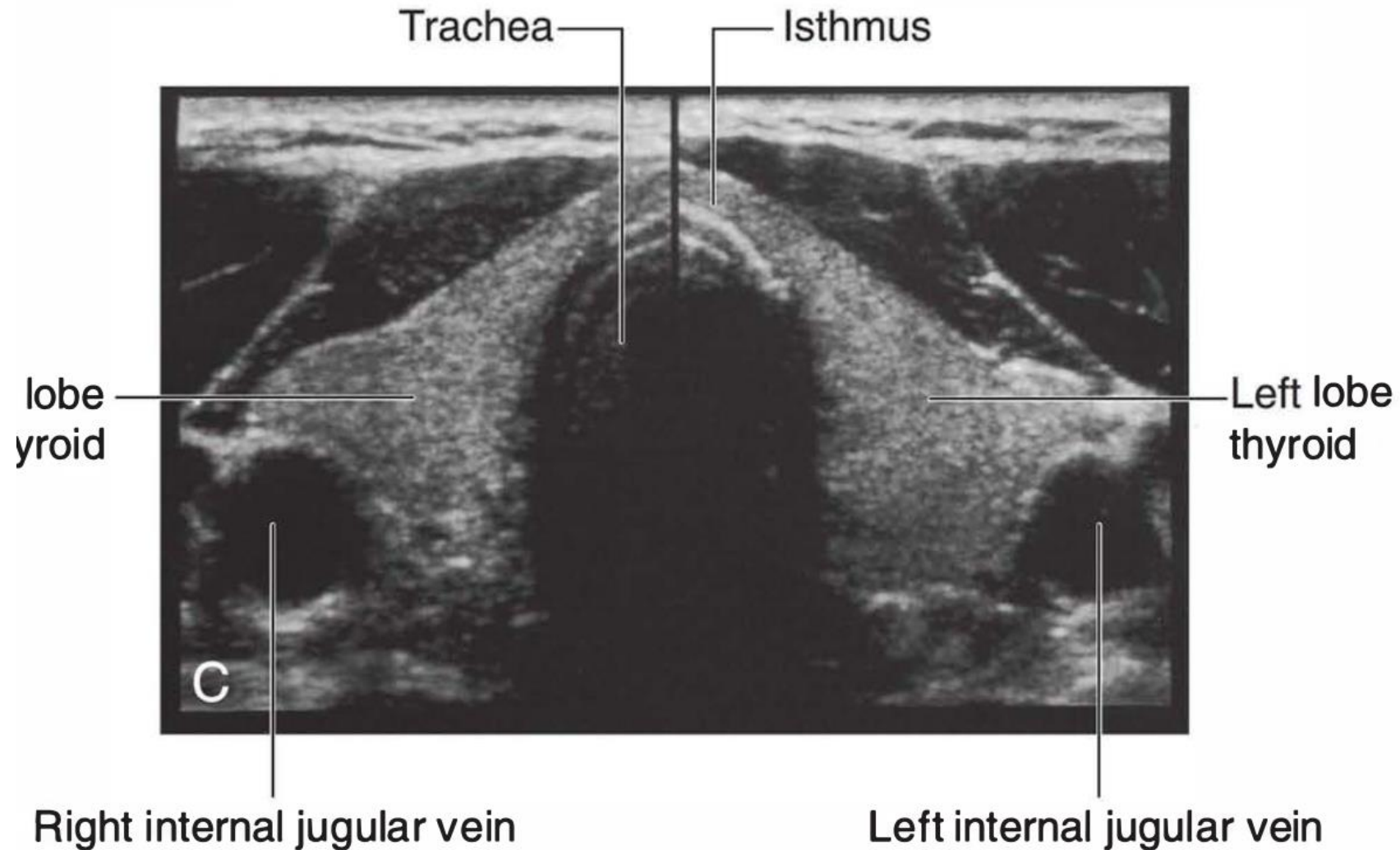
- Hyperthyroid
- T_3 High
- T_4 High
- TSH Low



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- Hypothyroidism
- T_3 Low
- T_4 Low
- TSH High

Ultrasound of the neck







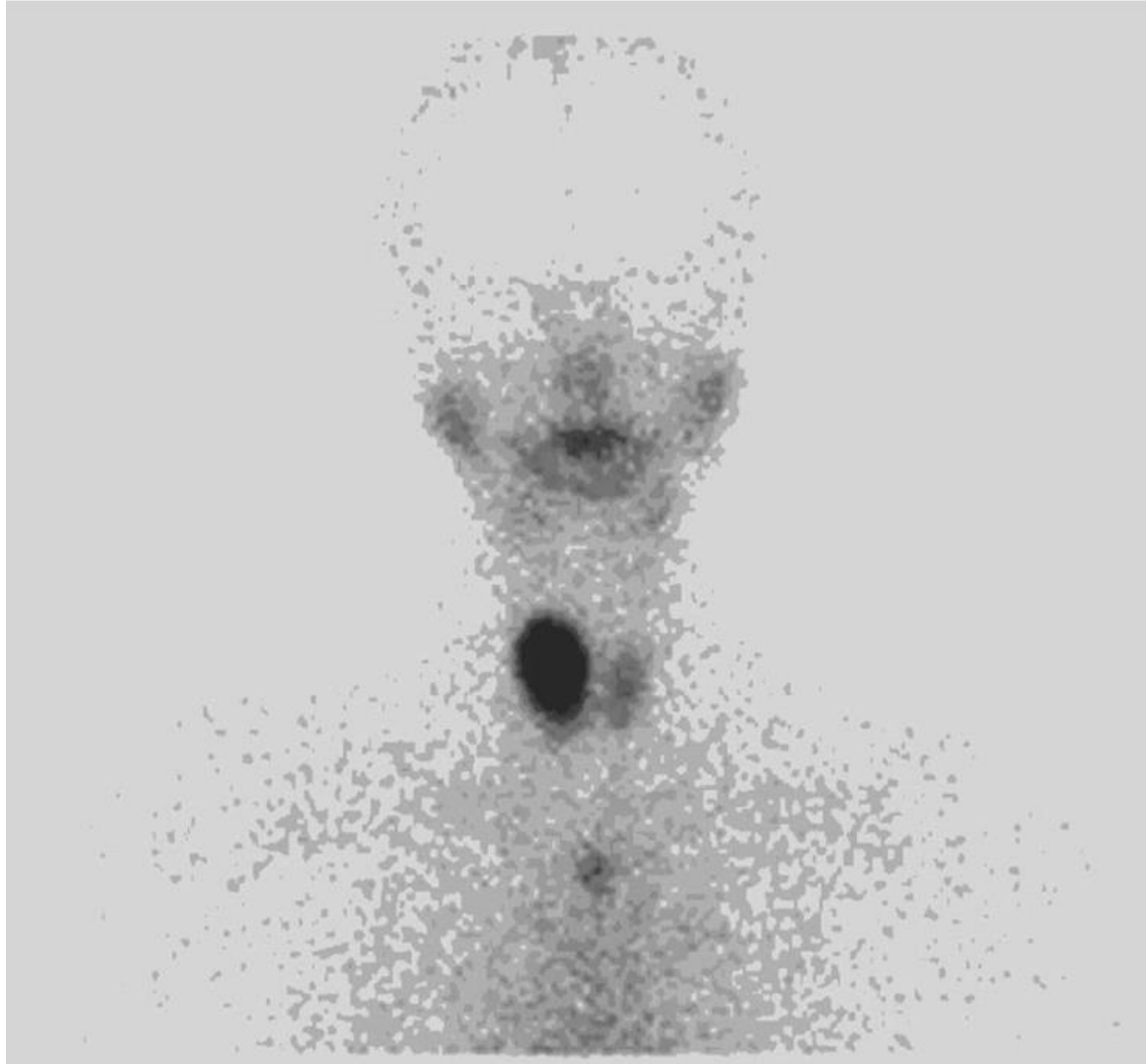
FNAC

- FNAC thyroid, or fine needle aspiration cytology of the thyroid, is a minimally invasive procedure that is used to diagnose thyroid nodules.
- Once the needle is in place, suction will be applied to withdraw a sample of cells.
- The cells are then placed on a slide and sent to a laboratory for analysis.

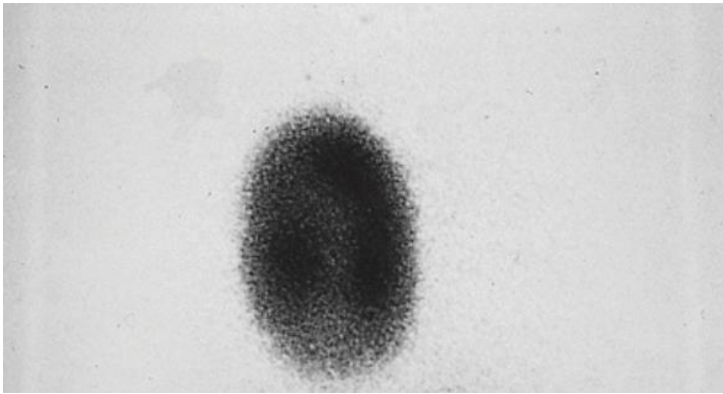




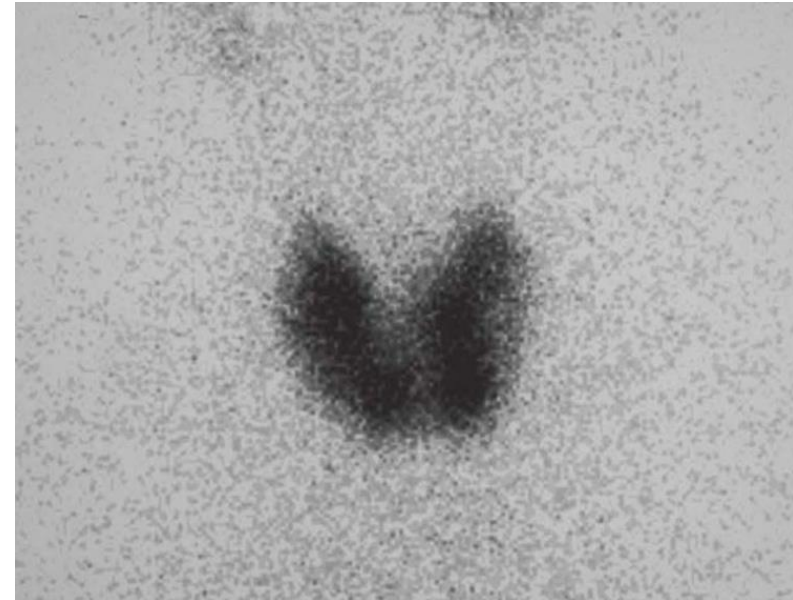
Thyroid scan



- The thyroid scan is a nuclear medicine imaging test that uses Tc99m to create images of the thyroid gland.



Toxic adenoma in the right thyroid was demonstrated with the use of technetium-pertechnetate scanning.







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Thyroid toxic crisis (thyroid storm)

It's a life-threatening complication of hyperthyroidism.
It occurs when the thyroid gland suddenly releases a large amount of thyroid hormone into the bloodstream.



Causes

- Untreated or poorly controlled hyperthyroidism
- Stress
- Infection.
- Trauma
- Certain medications



- Symptoms of thyroid toxic crisis include:
- High fever
- Rapid heartbeat
- Agitation and confusion
- Diarrhea and vomiting
- Muscle weakness
- Coma



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Treatment

- Admission to hospital
- Beta-blockers
- Antithyroid medications
- Iodine preparations
- Glucocorticoids



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Hypothyroidism

- Patients with hypothyroidism should be identified when possible because medical treatment can greatly improve their quality of life.



- With detection early in childhood, permanent mental retardation can be avoided with appropriate medical management.
- In addition, oral complications of delayed eruption of teeth, malocclusion, enlarged tongue, and skeletal growth impairment can be prevented through early detection and medical treatment.



Common thyroidectomy complications

- Hypocalcemia (low calcium): This is the most common complication. managed with calcium and vitamin D supplements.
- Hoarseness: Damage to the recurrent laryngeal nerve can lead to hoarseness or voice changes.
- This often improves within a few weeks to months



- Bleeding or hematoma can cause swelling, pain, and pressure, and may require emergency surgery to relieve pressure on the airway.



Less common complications

- Injury to the superior laryngeal nerve, which innervates the cricothyroid muscle, can affect voice quality.
- Infection



- Esophageal or tracheal injury: These are rare but serious complications that may require immediate repair.
- Thyroid storm.



- Hypothyroidism: If the entire thyroid is removed and not replaced with hormone medication, you will develop hypothyroidism (underactive thyroid).
- Seroma



- Surgical emphysema: Air can get trapped under the skin, which is more common after certain types of transoral thyroidectomy.



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Thank you